Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

***Ringkasan*—This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60**

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

url [colorlinks=false]hyperref
[pdftex]graphicx
amsmath bm
siunitx
array
booktabs multirow

I. Introduction

Use this LaTeX template to document your homework assignment solution. The template is based on the IEEE journal format standard, which is widespread for scientific papers in the engineering community. If you are not an advanced LaTeX user, do not be afraid since all relevant commands needed to insert your solutions in this document will be briefly explained next.

A. Equations

You can insert equations with numbering

$$a^2 + b^2 = c^2 \tag{1}$$

or without

$$c = \pm\sqrt{a^2 + b^2}.$$

Multi line equations are also possible

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = Ax(t) + Bu(t), y(t) = Cx(t) + Du(t). \tag{2}$$

Here, vectors and matrices have been depicted in bold font using the `\bm{}` command in math mode. Although many symbols in mathematical formulae are depicted in italic font, on the contrary, some require upright font such as

- mathematical operators (e.g. the differential $\mathrm{d}/\mathrm{d}t$),
- unit symbols such as ampere, volt, seconds, newton, etc. (e.g., 1 = 1),
- function names like $\cos, \sin$ and
- descriptive sub and superscripts (e.g. x_{start}).

In this context, the `\mathrm{}` command can be very helpful. Equations can be also inserted within the text lines (e.g., $x = y$) using the dollar sign notation `$x=y$`. You can also reference equations, e.g. eq:arbitrary$_l abel, by using the$ `\eqref{}` $command which automatically adjusts the equation reference number in case the equ$

Additional resources on mathematical representations:

- List of most common math commands and symbols: https://en.wikibooks.org/wiki/LaTeX/Mathematics
- Typographical conventions: https://en.wikipedia.org/wiki/Typographical_conventions_in_mathematical_formulae

B. Figures

Figures and plots can be inserted using the `\includegraphics[]{}` command which is normally encapsulated within a floating area defined by `\begin{figure}\end{figure}`. Likewise equations, figures can be labeled and referenced automatically within the document, cf. Fig. 1.

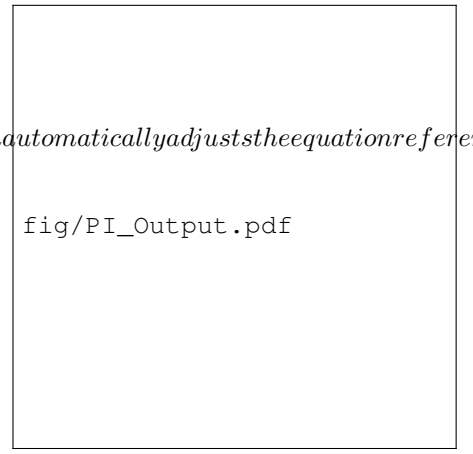
In engineering documents many figures are representing technical information such as block diagrams (cf. Fig. 2) or results from simulations (cf. Fig. 3). In such cases it is highly recommended to use vector graphic formats which are normally embedded within pdf files. Compared to pixel-based graphics the resulting figure quality is crystal clear and requires (in most cases) much less data space.

Additional resources on figures:

- Some general information on importing graphics in LaTeX: https://en.wikibooks.org/wiki/LaTeX/Importing_Graphics



Gambar 1. Who knows that guy?



Gambar 2. Example of a block diagram (vector graphic)



Gambar 3. Example of a simulation result (also vector graphic)

- Exporting figures from Matlab with `print()`: https://www.mathworks.com/help/matlab/ref/print.html
- External toolbox for Matlab figure exporting: https://github.com/altmany/export_fig
- For drawing own vector graphics (e.g. for block diagrams) the open-source software Inkscape (https://inkscape.org/) is highly recommended.
- With TikZ (https://texample.net/tikz/examples/) exists also an alternative to "draw" figures within LaTeX. But it requires some time to get used to command line drawing.

C. Tables

Tables are also encapsulated in floats called via `\begin{table}\end{table}` by default. An example can be found in Tab. I. There it should be noted that vertical lines are avoided – which is a very good rule of thumb to create nice, easy readable tables. Additional resources:

- Background information on table types and definitions: https://de.overleaf.com/learn/latex/tables
- Particular nice tables with `booktabs`: https://nhigham.com/2019/11/19/better-latex-tables-with-booktabs/

Tabel I
FANCY TABLE

Models	precision	$\|x\|_2$	$\|x\|_1$	$\|x\|_\infty$
model 1	0.67	0.8	0.729	0.75
model 2	0.8	0.9	0.847	0.85

D. Further specifications and hints

- In most scientific papers American English is used. Stick to this convention for your solution.
- Use concise, to the point sentences that guide the reader through the text. Most scientific publishers only accept papers of certain maximum length, so text space is precious. Also, nobody has time to read lengthy documents.
- Avoid unobjective, emotional language – remain neutral in your evaluation and factual in your choice of words.
- Use Google (or another search engine of your choice): LaTeX is the default typesetting framework for scientific texts. If you have any question on how to do something within LaTeX or getting some compilation errors or searching for a beginners tutorial: you will find it with a simple internet research in no time.
- You can write your solution in your free-of-charge Overleaf account or you can download the template and install a local LaTeX distribution (cf. https://www.latex-project.org/get/) on your personal machine. In either case, just submit the final pdf as your homework solution and not the source files.
- Be independent and get to the bottom of things. When starting to work in industry or academia your boss wants to see a solution solver and not problem deliverer (which includes writing nice texts in LaTeX).

II. TASK 1

Your solution goes here

III. TASK 2

Your solution goes here

IV. TASK ...

Your solution goes here